

Henri Heyden Mika Materzok Nike Pulow Senanur Şimşek

– Billy Joel

Not only is it interesting to look at popular music from the point of view of musicians aspiring to create the next hit single. Indeed, it is also interesting to analyse recent music since music is an essential part of any culture. By analysing popular music one can learn about what the current culture is, how it changed and how it varied. In the following, we'll look at the most popular music of the last two decades [1] to answer which musical and lyrical elements we can identify and analyse the change of them.

What main chord repetition should you use to write the perfect song? By webscraping *Ultimate Guitar* [7], we found the most frequently used harmonic progressions of the last 20 years, see figure 1.

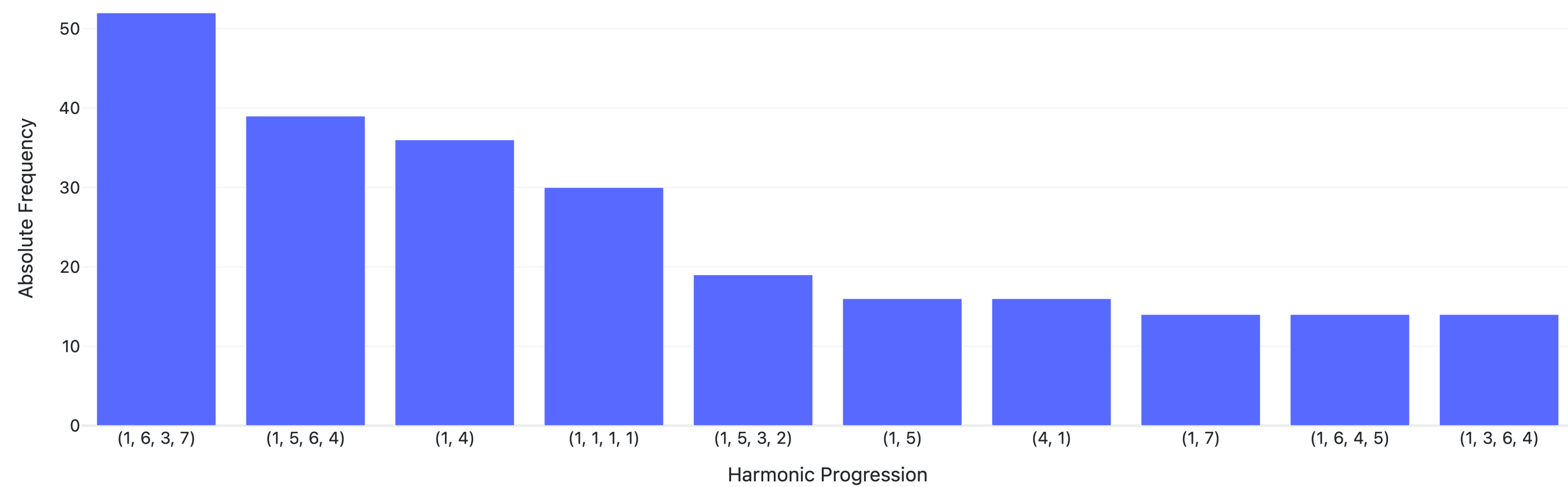


Figure 1: Top ten most used harmonic progressions by absolute frequency

Musicians often write songs in the same key they can sing in and therefore use the same chords more often. Most chords in figure 2 can be played in C-major or A-minor. We can also see that A-minor key is more popular than C-major key, since D and Em work better in A-minor compared to C-major.¹

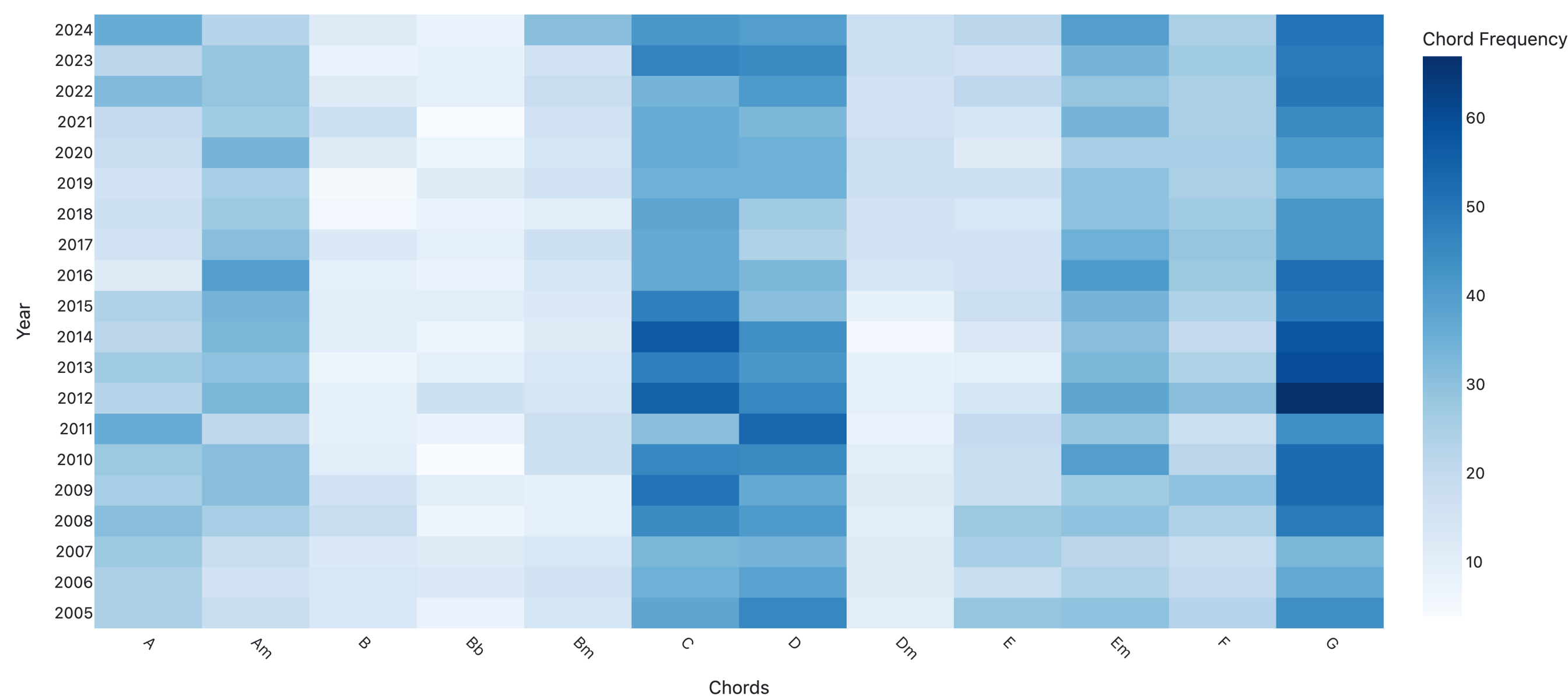


Figure 2: Top ten most used basic chords over the years

It shouldn't be surprising, that the most frequent genre in popular music is *pop*-music. Looking at the top tags of the song we got from the *LastFM API* [3] and comparing to the identified harmonic progressions, we can see which genres use mainly which chord progressions, see figure 3. Here we can identify, how pop dominates in more complex progressions compared to rap and hip-hop.

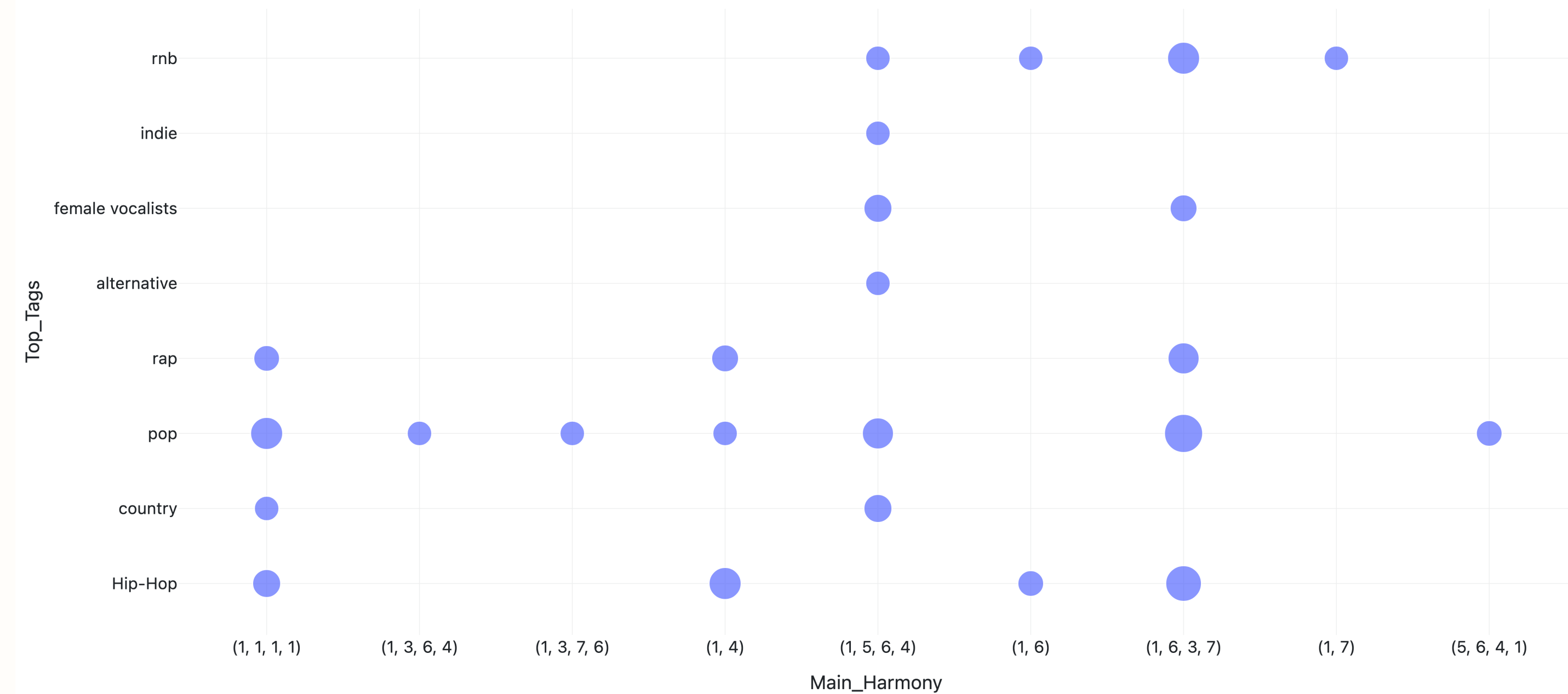


Figure 3: frequent genre tags over how often they use a chord progression with less frequent combinations filtered out

¹D and Em are roughly the IV and V of A-minor, but the II and III of C-major, which are less important harmonics, as you can see in figure 1.

Song lyrics often reflect emotions and moods of their time. By extracting them from *Genius API* [2] and analyzing their polarity over the years with *TextBlob* [6], we can identify cultural trends about how the language in songs had become positively or negatively charged, see figure 4.

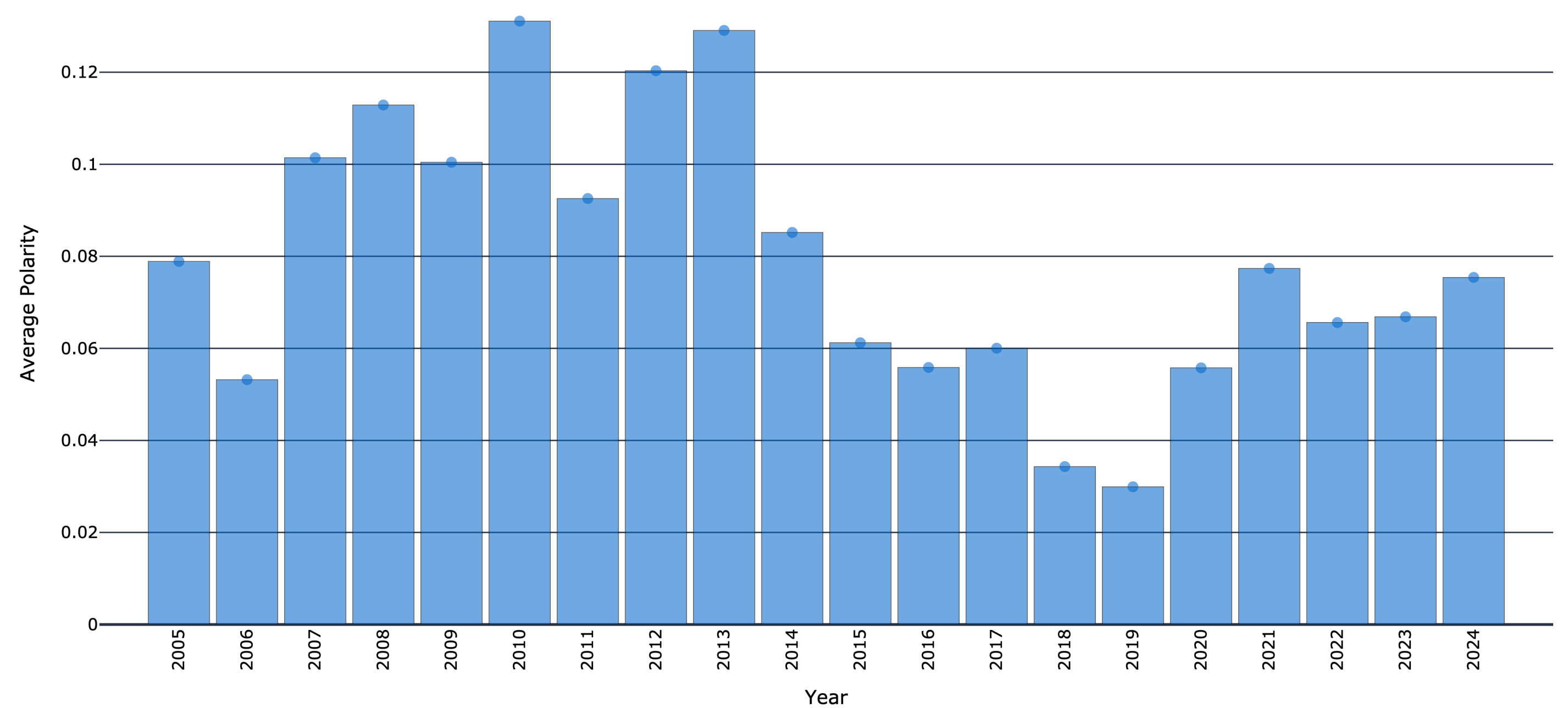


Figure 4: Song lyric polarity over time

In addition to polarity, the most frequently used words in song lyrics provide insight into lyrical trends. Figure 5 displays the most common words appearing in lyrics over the years.



Figure 5: Wordcloud of most frequent and important words of all lyrics

The duration of songs has evolved over time, reflecting changes in listening habits and industry standards. We received the duration of songs with *Spotify API* [5]. Even though minimal, we can see a small general trend to shorter songs in figure 6. A length of 3 minutes and 30 seconds seems ideal.

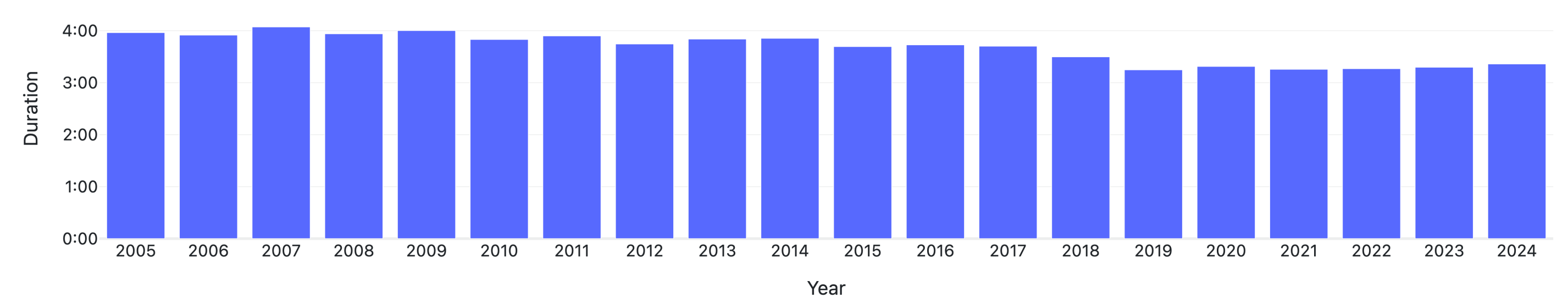


Figure 6: Average song duration over the years

Tempo plays a crucial role in defining the energy and feel of a song. We used *SongBPM* [4] to find the beats per minute of the songs. In figure 7 we can see how it changed and how variant it is.

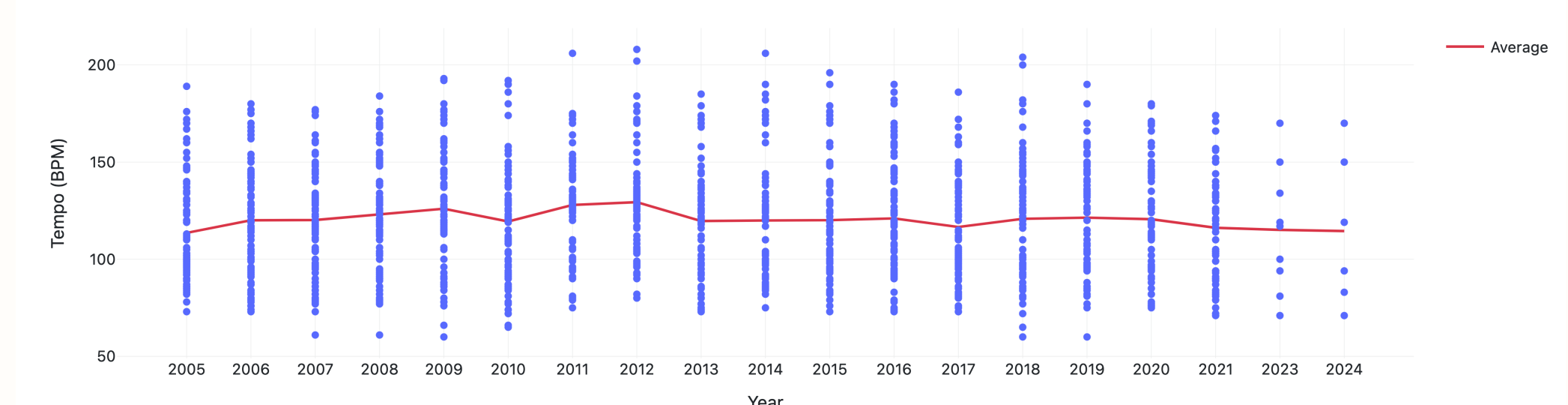


Figure 7: Average song tempo in beats per minute over the years

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